Chapter 23: Problems with solutions

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4. A charged particle A exerts a force of 2.62 μN to the right on charged particle B when the particles are 13.7 mm apart. Particle B moves straight away from A to make the distance between them 17.7 mm. What vector force does it then exert on A?

In the first situation, $\vec{\mathbf{F}}_{A \text{ on } B,1} = \dfrac{k_e |q_A||q_B|}{r_1^2}\hat{\mathbf{i}}$. In the second situation, $|q_A|$ and $|q_B|$ are the same.

$$\vec{\mathbf{F}}_{B \text{ on } A,2} = -\vec{\mathbf{F}}_{A \text{ on } B} = \frac{k_e |q_A||q_B|}{r_2^2}\left(-\hat{\mathbf{i}}\right)$$

$$\frac{F_2}{F_1} = \frac{k_e |q_A||q_B|}{r_2^2}\frac{r_1^2}{k_e |q_A||q_B|}$$

$$F_2 = \frac{F_1 r_1^2}{r_2^2} = (2.62\ \mu\text{N})\left(\frac{13.7\ \text{mm}}{17.7\ \text{mm}}\right)^2 = 1.57\ \mu\text{N}$$

Then $\vec{\mathbf{F}}_{B \text{ on } A,2} =$ **1.57 μN to the left**.

5. In a thundercloud, there may be electric charges of +40.0 C near the top of the cloud and −40.0 C near the bottom of the cloud. These charges are separated by 2.00 km. What is the electric force on the top charge?

The electric force is given by

$$F = k_e \frac{q_1 q_2}{(r_{12})^2} = \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right) \frac{(+40 \text{ C})(-40 \text{ C})}{(2\,000 \text{ m})^2}$$

$$= -3.60 \times 10^6 \text{ N (attractive)} = \boxed{3.60 \times 10^6 \text{ N downward}}$$

6. (a) Find the magnitude of the electric force between a Na^+ ion and a Cl^- ion separated by 0.50 nm. (b) Would the answer change if the sodium ion were replaced by Li^+ and the chloride ion by Br^-? Explain.

(a) The two ions are both singly charged, $|q| = 1e$, one positive and one negative. Thus,

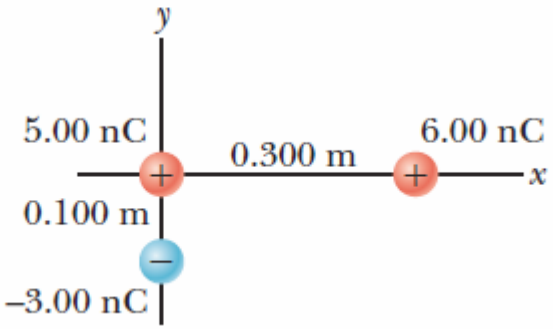
$$|F| = \frac{k_e |q_1||q_2|}{r^2} = \frac{k_e e^2}{r^2}$$

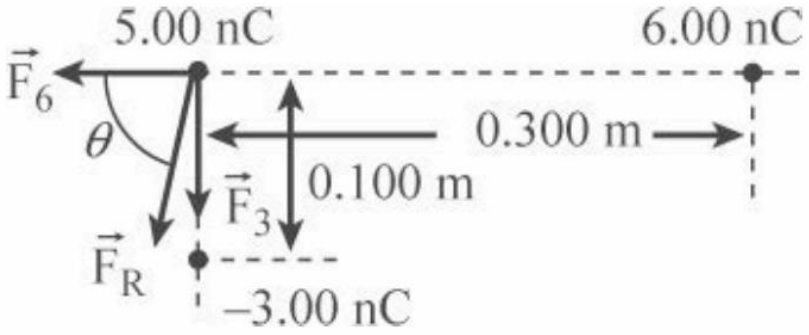
$$= \frac{\left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right)\left(1.60 \times 10^{-19} \text{ C}\right)^2}{\left(0.500 \times 10^{-9} \text{ m}\right)^2}$$

$$= \boxed{9.21 \times 10^{-10} \text{ N}}$$

(b) No. The electric force depends only on the magnitudes of the two charges and the distance between them.

11. M Three point charges are arranged as shown in Figure P23.11. Find (a) the magnitude and (b) the direction of the electric force on the particle at the origin.





$$F_x = -F_6 = -\left(8.99\times10^{9}\ \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}\right)\frac{(6.00\times10^{-9}\ \text{C})(5.00\times10^{-9}\ \text{C})}{(0.300\ \text{m})^2}$$

$$= -3.00\times10^{-6}\ \text{N} \quad \text{(to the left)}$$

$$F_y = -F_3 = -\left(8.99\times10^{9}\ \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}\right)\frac{(3.00\times10^{-9}\ \text{C})(5.00\times10^{-9}\ \text{C})}{(0.100\ \text{m})^2}$$

$$= -1.35\times10^{-5}\ \text{N} \quad \text{(downward)}$$

(a) The forces are perpendicular, so the magnitude of the resultant is

$$F_R = \sqrt{(F_6)^2 + (F_3)^2} = \boxed{1.38\times10^{-5}\ \text{N}}$$

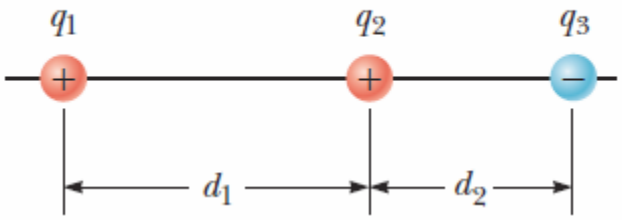
(b) The magnitude of the angle of the resultant is

$$\theta = \tan^{-1}\left(\frac{F_3}{F_6}\right) = 77.5°$$

The resultant force is in the third quadrant, so the direction is

$$\boxed{77.5°\ \text{below}\ -x\ \text{axis}}$$

12. Three point charges lie along a straight line as shown in Figure P23.12, where $q_1 = 6.00\ \mu\text{C}$, $q_2 = 1.50\ \mu\text{C}$, and $q_3 = -2.00\ \mu\text{C}$. The separation distances are $d_1 = 3.00$ cm and $d_2 = 2.00$ cm. Calculate the magnitude and direction of the net electric force on (a) q_1, (b) q_2, and (c) q_3.



$$F_1 = \frac{k_e q_1 q_2}{r_{12}^2} = \left(8.99\times10^{9}\ \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}\right)\frac{(6.00\times10^{-6}\ \text{C})(1.50\times10^{-6}\ \text{C})}{(3.00\times10^{-2}\ \text{m})^2}$$

$$= 89.9\ \text{N}$$

$$F_2 = \frac{k_e q_1 |q_3|}{r_{13}^2} = \left(8.99\times10^{9}\ \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}\right)\frac{(6.00\times10^{-6}\ \text{C})(2.00\times10^{-6}\ \text{C})}{(5.00\times10^{-2}\ \text{m})^2}$$

$$= 43.2\ \text{N}$$

$$F_3 = \frac{k_e q_2 |q_3|}{r_{23}^2} = \left(8.99\times10^{9}\ \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}\right)\frac{(1.50\times10^{-6}\ \text{C})(2.00\times10^{-6}\ \text{C})}{(2.00\times10^{-2}\ \text{m})^2}$$

$$= 67.4\ \text{N}$$

(a) The net force on the 6 μC charge is

$$F_{(6\mu C)} = F_1 - F_2 = \boxed{46.7 \text{ N to the left}}$$

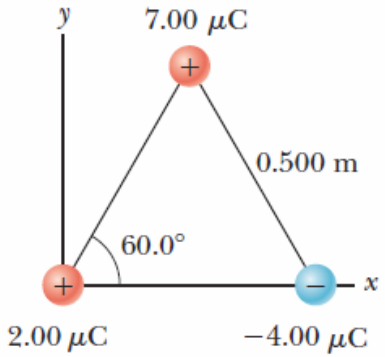
(b) The net force on the 1.5 μC charge is

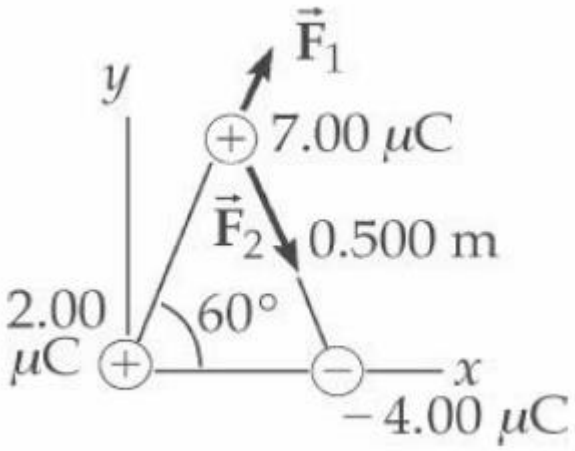
$$F_{(1.5\mu C)} = F_1 + F_3 = \boxed{157 \text{ N to the right}}$$

(c) The net force on the –2 μC charge is

$$F_{(-2\mu C)} = F_2 + F_3 = \boxed{111 \text{ N to the left}}$$

15. M Three charged particles are located at the corners of an equilateral triangle as shown in Figure P23.15. Calculate the total electric force on the 7.00-μC charge.





The force exerted on the 7.00-μC charge by the 2.00-μC charge is

$$\vec{\mathbf{F}}_1 = k_e \frac{q_1 q_2}{r^2}\hat{\mathbf{r}}$$

$$= \frac{(8.99\times10^{9}\ \mathrm{N\cdot m^2/C^2})(7.00\times10^{-6}\ \mathrm{C})(2.00\times10^{-6}\ \mathrm{C})}{(0.500\ \mathrm{m})^2}$$

$$\times(\cos 60^\circ \hat{\mathbf{i}} + \sin 60^\circ \hat{\mathbf{j}})$$

$$\vec{\mathbf{F}}_1 = (0.252\ \hat{\mathbf{i}} + 0.436\ \hat{\mathbf{j}})\ \mathrm{N}$$

Similarly, the force on the 7.00-μC charge by the –4.00-μC charge is

$$\mathbf{F}_2 = k_e \frac{q_1 q_3}{r^2}\hat{\mathbf{r}}$$

$$= -\frac{(8.99\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(7.00\times10^{-6}\ \text{C})(-4.00\times10^{-6}\ \text{C})}{(0.500\ \text{m})^2}$$

$$\times(\cos 60^\circ \hat{\mathbf{i}} - \sin 60^\circ \hat{\mathbf{j}})$$

$$\vec{\mathbf{F}}_2 = (0.503\hat{\mathbf{i}} - 0.872\ \hat{\mathbf{j}})\ \text{N}$$

Thus, the total force on the 7.00-μC charge is

$$\vec{\mathbf{F}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 = (0.755\ \hat{\mathbf{i}} - 0.436\ \hat{\mathbf{j}})\ \text{N}$$

We can also write the total force as:

$$\vec{\mathbf{F}} = (0.755\ \text{N})\hat{\mathbf{i}} - (0.436\ \text{N})\hat{\mathbf{j}} = \boxed{0.872\ \text{N at an angle of } 330^\circ}$$

16. Two small metallic spheres, each of mass $m = 0.200$ g, are suspended as pendulums by light strings of length L as shown in Figure P23.16. The spheres are given the same electric charge of 7.2 nC, and they come to equilibrium when each string is at an angle of $\theta = 5.00°$ with the vertical. How long are the strings?

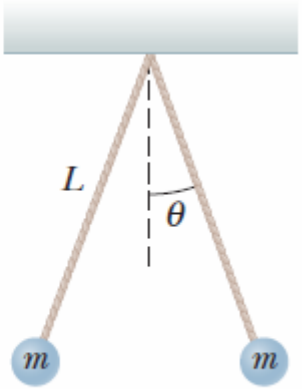
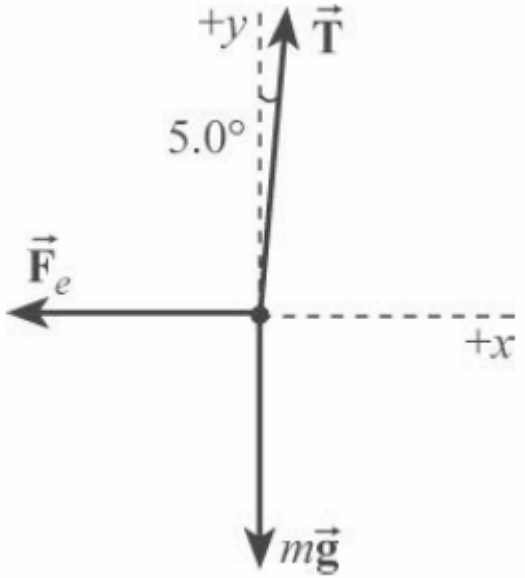


Figure P23.16



$$\sum F_y = 0 \Rightarrow T\cos 5.0° = mg$$

or $$T = \frac{mg}{\cos 5.0°}$$

$$\sum F_x = 0 \Rightarrow F_e = T\sin 5.0° = mg\tan 5.0°$$

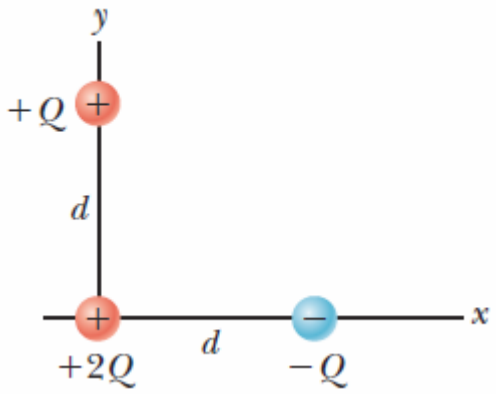
At equilibrium, the distance separating the two spheres is $r = 2L\sin 5.0°$.

Thus, $F_e = mg\tan 5.0°$ becomes $\dfrac{k_e q^2}{(2L\sin 5.0°)^2} = mg\tan 5.0°$, which yields

$$L = \sqrt{\frac{k_e q^2}{mg\tan 5.0°\,(2\sin 5.0°)^2}}$$

$$= \sqrt{\frac{(8.99\times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(7.20\times 10^{-9}\ \text{C})^2}{(0.200\times 10^{-3}\ \text{kg})(9.80\ \text{m/s}^2)\tan 5.0°\,(2\sin 5.0°)^2}} = \boxed{0.299\ \text{m}}$$

19. A point charge $+2Q$ is at the origin and a point charge $-Q$ is located along the x axis at $x = d$ as in Figure P23.19. Find a symbolic expression for the net force on a third point charge $+Q$ located along the y axis at $y = d$.

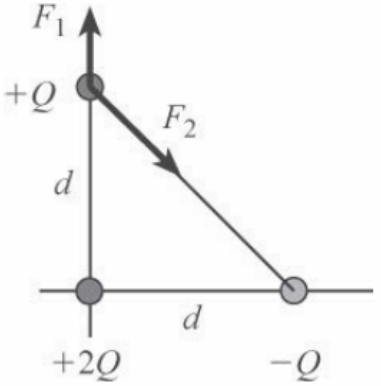


The force due to the first charge is given by

$$\vec{\mathbf{F}}_1 = \frac{k_e Q(2Q)}{d^2}\hat{\mathbf{j}} = \frac{k_e Q^2}{d^2}\left[2\hat{\mathbf{j}}\right]$$

and the force due to the second charge is given by

$$\vec{\mathbf{F}}_2 = \frac{k_e Q(Q)}{\left(d^2 + d^2\right)}\left[\frac{\hat{\mathbf{i}} - \hat{\mathbf{j}}}{\sqrt{2}}\right] = \frac{k_e Q^2}{d^2}\left[\frac{\hat{\mathbf{i}} - \hat{\mathbf{j}}}{2\sqrt{2}}\right]$$



ANS. FIG. P23.19

thus the total force on the point charge +Q located at $x = 0$ and $y = d$ is

$$\vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 = \frac{k_e Q^2}{d^2}\left[2\hat{\mathbf{j}}\right] + \frac{k_e Q^2}{d^2}\left[\frac{\hat{\mathbf{i}} - \hat{\mathbf{j}}}{2\sqrt{2}}\right] = \boxed{k_e \frac{Q^2}{d^2}\left[\frac{1}{2\sqrt{2}}\hat{\mathbf{i}} + \left(2 - \frac{1}{2\sqrt{2}}\right)\hat{\mathbf{j}}\right]}$$

21. W Two identical conducting small spheres are placed with their centers 0.300 m apart. One is given a charge of 12.0 nC and the other a charge of −18.0 nC. (a) Find the electric force exerted by one sphere on the other. (b) **What If?** The spheres are connected by a conducting wire. Find the electric force each exerts on the other after they have come to equilibrium.

(a) The force is one of attraction. The distance r in Coulomb's law is the distance between the centers of the spheres. The magnitude of the force is

$$F = \frac{k_e q_1 q_2}{r^2}$$

$$= \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2\right) \frac{\left(12.0 \times 10^{-9} \text{ C}\right)\left(18.0 \times 10^{-9} \text{ C}\right)}{(0.300 \text{ m})^2}$$

$$= \boxed{2.16 \times 10^{-5} \text{ N}}$$

(b) The net charge of -6.00×10^{-9} C will be equally split between the two spheres, or -3.00×10^{-9} C on each. The force is one of repulsion, and its magnitude is

$$F = \frac{k_e q_1 q_2}{r^2}$$

$$= \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2\right) \frac{\left(3.00 \times 10^{-9} \text{ C}\right)\left(3.00 \times 10^{-9} \text{ C}\right)}{(0.300 \text{ m})^2}$$

$$= \boxed{8.99 \times 10^{-7} \text{ N}}$$

23. What are the magnitude and direction of the electric field that will balance the weight of (a) an electron and (b) a proton? (You may use the data in Table 23.1.)

For equilibrium, $\vec{\mathbf{F}}_e = -\vec{\mathbf{F}}_g$ or $q\vec{\mathbf{E}} = -mg\left(-\hat{\mathbf{j}}\right)$. Thus,

$$\vec{\mathbf{E}} = \frac{mg}{q}\hat{\mathbf{j}}.$$

(a) For an electron,

$$\vec{\mathbf{E}} = \frac{mg}{q}\hat{\mathbf{j}} = \frac{\left(9.11\times10^{-31}\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right)}{-1.60\times10^{-19}\ \text{C}}\hat{\mathbf{j}}$$

$$= \boxed{-\left(5.58\times10^{-11}\ \text{N/C}\right)\hat{\mathbf{j}}}$$

(b) For a proton, which is 1 836 times more massive than an electron,

$$\vec{\mathbf{E}} = \frac{mg}{q}\hat{\mathbf{j}} = \frac{\left(1.67\times10^{-27}\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right)}{-1.60\times10^{-19}\ \text{C}}\hat{\mathbf{j}}$$

$$= \boxed{\left(1.02\times10^{-7}\ \text{N/C}\right)\hat{\mathbf{j}}}$$

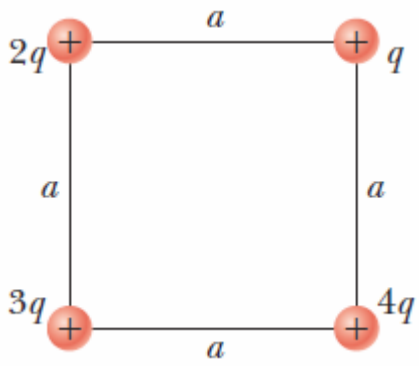
24. A small object of mass 3.80 g and charge $-18.0\ \mu C$ is suspended motionless above the ground when immersed in a uniform electric field perpendicular to the ground. What are the magnitude and direction of the electric field?

In order for the object to "float" in the electric field, the electric force exerted on the object by the field must be directed upward and have a magnitude equal to the weight of the object. Thus, $F_e = qE = mg$, and the magnitude of the electric field must be

$$E = \frac{mg}{|q|} = \frac{(3.80 \times 10^{-3}\ \text{kg})(9.80\ \text{m/s}^2)}{18.0 \times 10^{-6}\ \text{C}} = \boxed{2.07 \times 10^{3}\ \text{N/C}}$$

The electric force on a negatively charged object is in the direction opposite to that of the electric field. Since the electric force must be directed upward, the electric field must be directed $\boxed{\text{downward}}$.

25. Four charged particles are at the corners of a square of side a as shown in Figure P23.25. Determine (a) the electric field at the location of charge q and (b) the total electric force exerted on q.



We sum the electric fields from each of the other charges using Equation 23.7 for the definition of the electric field.

The field at charge q is given by

$$\vec{\mathbf{E}} = \frac{k_e q_1}{r_1^2}\hat{\mathbf{r}}_1 + \frac{k_e q_2}{r_2^2}\hat{\mathbf{r}}_2 + \frac{k_e q_3}{r_3^2}\hat{\mathbf{r}}_3$$

(a) Substituting for each of the charges gives

$$\vec{\mathbf{E}} = \frac{k_e (2q)}{a^2}\hat{\mathbf{i}} + \frac{k_e (3q)}{2a^2}\left(\hat{\mathbf{i}}\cos 45.0^\circ + \hat{\mathbf{j}}\sin 45.0^\circ\right) + \frac{k_e (4q)}{a^2}\hat{\mathbf{j}}$$

$$= \frac{k_e q}{a^2}\left[\left(2 + \frac{3}{2}\cos 45.0^\circ\right)\hat{\mathbf{i}} + \left(\frac{3}{2}\sin 45.0^\circ + 4\right)\hat{\mathbf{j}}\right]$$

$$= \boxed{\frac{k_e q}{a^2}\left(3.06\hat{\mathbf{i}} + 5.06\hat{\mathbf{j}}\right)}$$

(b) The electric force on charge q is given by

$$\vec{\mathbf{F}} = q\vec{\mathbf{E}} = \boxed{\frac{k_e q^2}{a^2}\left(3.06\hat{\mathbf{i}} + 5.06\hat{\mathbf{j}}\right)}$$

29. In Figure P23.29, determine the point (other than infinity) at which the electric field is zero.

M

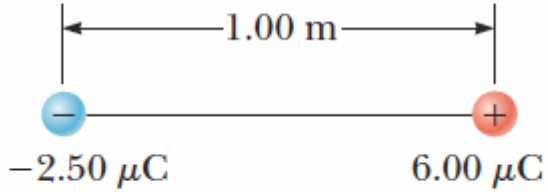
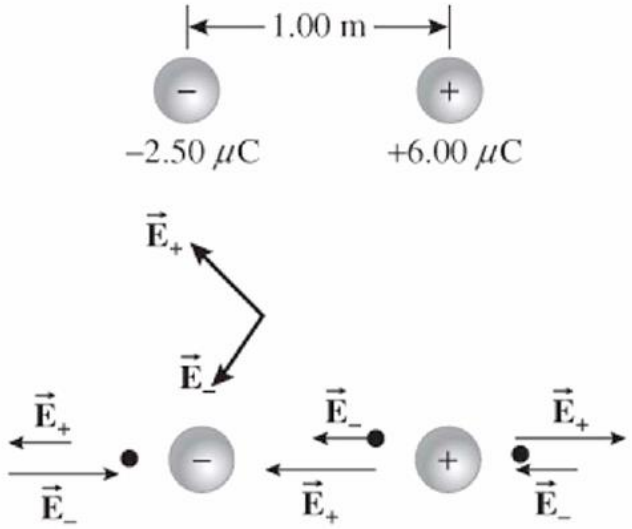


Figure P23.29



$$\frac{k_e|q_-|}{x^2} = \frac{k_e|q_+|}{(1\text{ m} + x)^2}$$

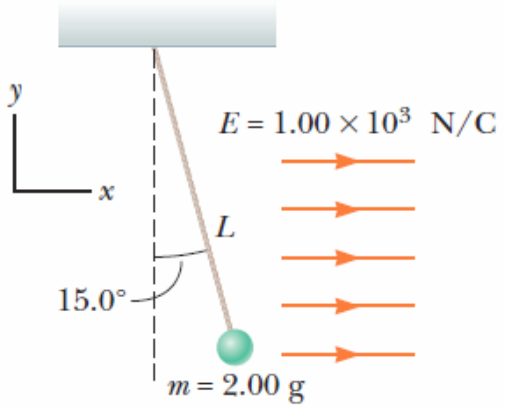
$$|q_-|^{1/2}(1\text{ m} + x) = q_+^{1/2}x$$

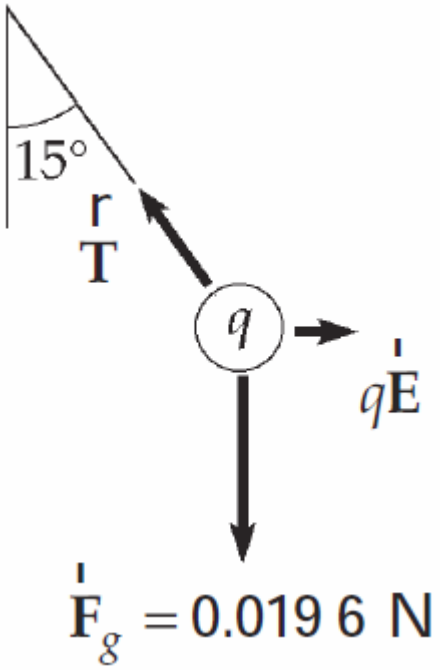
$$1\text{ m} + x = \left(\frac{6.00}{2.50}\right)^{1/2} x = 1.55\,x$$

$$1\text{ m} = 0.549x$$

and $x = \boxed{1.82\text{ m}}$ to the left of the negatively-charged object.

33. AMT A small, 2.00-g plastic ball is suspended by a 20.0-cm-long string in a uniform electric field as shown in Figure P23.33. If the ball is in equilibrium when the string makes a 15.0° angle with the vertical, what is the net charge on the ball?





From the free-body diagram shown in ANS. FIG. P23.33,

$$\sum F_y = 0: \qquad T\cos 15.0° = 1.96 \times 10^{-2}\ \text{N}$$

So $T = 2.03 \times 10^{-2}$ N.

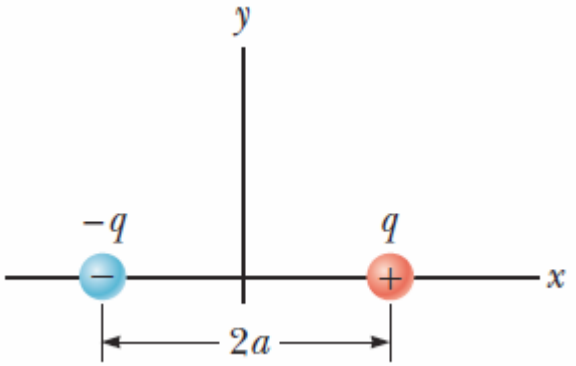
From $\sum F_x = 0$, we have $qE = T\sin 15.0°$,

or

$$q = \frac{T\sin 15.0°}{E} = \frac{(2.03 \times 10^{-2}\ \text{N})\sin 15.0°}{1.00 \times 10^3\ \text{N/C}}$$

$$= 5.25 \times 10^{-6}\ \text{C} = \boxed{5.25\ \mu\text{C}}$$

36. Consider the electric dipole shown in Figure P23.36. Show that the electric field at a *distant* point on the $+x$ axis is $E_x \approx 4k_e qa/x^3$.



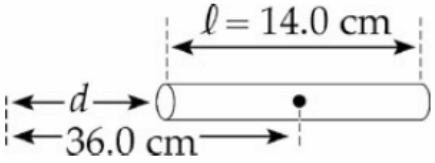
The electric field at any point x is

$$E = \frac{k_e q}{(x-a)^2} - \frac{k_e q}{[x-(-a)]^2} = \frac{k_e q(4ax)}{\left(x^2-a^2\right)^2}$$

When x is much, much greater than a, we find $E \approx \boxed{\dfrac{4a(k_e q)}{x^3}}$.

37. A rod 14.0 cm long is uniformly charged and has a total charge of $-22.0\ \mu C$. Determine (a) the magnitude and (b) the direction of the electric field along the axis of the rod at a point 36.0 cm from its center.

(a) From Example 23.7, the magnitude of the electric field produced by the rod is



ANS. FIG. P23.37

$$|E| = \frac{k_e \lambda \ell}{d(\ell + d)} = \frac{k_e (Q/\ell)\ell}{d(\ell + d)} = \frac{k_e Q}{d(\ell + d)}$$

$$= \frac{(8.99 \times 10^9\ \text{N} \cdot \text{m}^2/\text{C}^2)(22.0 \times 10^{-6}\ \text{C})}{(0.290\ \text{m})(0.140\ \text{m} + 0.290\ \text{m})}$$

$$E = \boxed{1.59 \times 10^6\ \text{N/C}}$$

(b) The charge is negative, so the electric field is directed towards its source, **to the right**.

38. A uniformly charged disk of radius 35.0 cm carries charge with a density of 7.90×10^{-3} C/m^2. Calculate the electric field on the axis of the disk at (a) 5.00 cm, (b) 10.0 cm, (c) 50.0 cm, and (d) 200 cm from the center of the disk.

The electric field for the disk is given by

$$E = 2\pi k_e \sigma \left(1 - \frac{x}{\sqrt{x^2 + R^2}} \right)$$

in the positive x direction (away from the disk). Substituting,

$$E = 2\pi \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2\right)\left(7.90 \times 10^{-3} \text{ C}/\text{m}^2\right) \times \left(1 - \frac{x}{\sqrt{x^2 + (0.350)^2}} \right)$$

$$= \left(4.46 \times 10^8 \text{ N}/\text{C}\right)\left(1 - \frac{x}{\sqrt{x^2 + 0.123}} \right)$$

(a) At $x = 0.050\ 0$ m,

$$E = 3.83 \times 10^{8}\ \text{N/C} = \boxed{383\ \text{MN/C}}$$

(b) At $x = 0.100$ m,

$$E = 3.24 \times 10^{8}\ \text{N/C} = \boxed{324\ \text{MN/C}}$$

(c) At $x = 0.500$ m,

$$E = 8.07 \times 10^{7}\ \text{N/C} = \boxed{80.7\ \text{MN/C}}$$

(d) At $x = 2.000$ m,

$$E = 6.68 \times 10^{8}\ \text{N/C} = \boxed{6.68\ \text{MN/C}}$$

39. M A uniformly charged ring of radius 10.0 cm has a total charge of 75.0 μC. Find the electric field on the axis of the ring at (a) 1.00 cm, (b) 5.00 cm, (c) 30.0 cm, and (d) 100 cm from the center of the ring.

We may particularize the result of Example 23.8 to

$$|E| = \frac{k_e x Q}{\left(x^2 + a^2\right)^{3/2}} = \frac{\left(8.99\times10^{9}\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(75.0\times10^{-6}\ \text{C/m}^2\right)x}{\left(x^2 + 0.100^2\right)^{3/2}}$$

$$= \frac{6.74\times10^{5}\,x}{\left(x^2 + 0.010\,0\right)^{3/2}}$$

where we choose the x axis along the axis of the ring. The field is parallel to the axis, directed away from the center of the ring above and below it.

(a) At $x = 0.010\,0$ m, $\vec{\mathbf{E}} = 6.64\times10^{6}\,\hat{\mathbf{i}}$ N/C = $\boxed{6.64\hat{\mathbf{i}}\ \text{MN/C}}$

(b) At $x = 0.050\,0$ m, $\vec{\mathbf{E}} = 2.41\times10^{7}\,\hat{\mathbf{i}}$ N/C = $\boxed{24.1\hat{\mathbf{i}}\ \text{MN/C}}$

(c) At $x = 0.300$ m, $\vec{\mathbf{E}} = 6.40\times10^{6}\,\hat{\mathbf{i}}$ N/C = $\boxed{6.40\hat{\mathbf{i}}\ \text{MN/C}}$

(d) At $x = 1.00$ m, $\vec{\mathbf{E}} = 6.64\times10^{5}\,\hat{\mathbf{i}}$ N/C = $\boxed{0.664\hat{\mathbf{i}}\ \text{MN/C}}$

43. A continuous line of charge lies along the x axis, **W** extending from $x = +x_0$ to positive infinity. The line carries positive charge with a uniform linear charge density λ_0. What are (a) the magnitude and (b) the direction of the electric field at the origin?

(a) Magnitude $|E| = \int \frac{k_e dq}{x^2}$, where $dq = \lambda_0 dx$

$$E = k_e \lambda_0 \int_{x_0}^{\infty} \frac{dx}{x^2} = k_e \lambda_0 \left(-\frac{1}{x} \right) \Bigg|_{x_0}^{\infty} = \boxed{\frac{k_e \lambda_0}{x_0}}$$

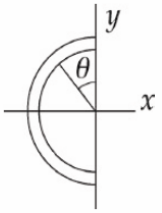
(b) The charge is positive, so the electric field points away from its source, to the left.

45. M A uniformly charged insulating rod of length 14.0 cm is bent into the shape of a semicircle as shown in Figure P23.45. The rod has a total charge of $-7.50\ \mu C$. Find (a) the magnitude and (b) the direction of the electric field at O, the center of the semicircle.



Due to symmetry, $E_y = \int dE_y = 0$, and

$E_x = -\int dE \sin\theta = -k_e \int \frac{dq \sin\theta}{r^2}$ where $dq = \lambda ds = \lambda r d\theta$; the component E_x is negative because charge $q = -7.50\ \mu C$, causing the net electric field to be directed to the left.



ANS. FIG. P23.45

$$E_x = -\frac{k_e \lambda}{r}\int_0^{\pi} \sin\theta d\theta = -\frac{k_e \lambda}{r}\left(-\cos\theta\right)\Big|_0^{\pi} = -\frac{2k_e \lambda}{r}$$

where $\lambda = \frac{|q|}{L}$ and $r = \frac{L}{\pi}$. Thus,

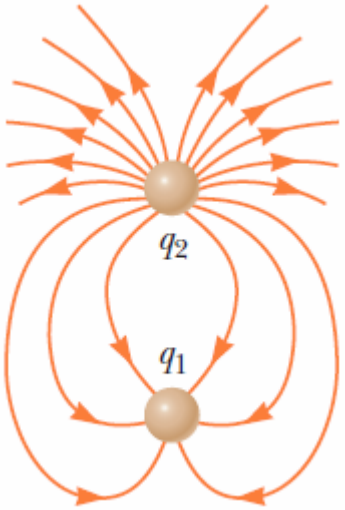
$$E_x = -\frac{2k_e |q| \pi}{L^2} = -\frac{2\left(8.99\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2\right)\left(7.50\times10^{-6}\ \text{C}\right)\pi}{(0.140\ \text{m})^2}$$

$$E_x = -2.16\times10^7\ \text{N/C}$$

(a) magnitude $E = \boxed{2.16\times10^7\ \text{N/C}}$

(b) **to the left**

49. Figure P23.49 shows the electric field lines for two charged particles separated by a small distance. (a) Determine the ratio q_1/q_2. (b) What are the signs of q_1 and q_2?



Field lines emerge from positive charge and enter negative charge.

(a) The number of field lines emerging from positive q_2 and entering negative charge q_1 is proportional to their charges:

$$\frac{q_1}{q_2} = \frac{-6}{18} = \boxed{-\frac{1}{3}}$$

(b) From above, $\boxed{q_1 \text{ is negative, } q_2 \text{ is positive}}$.

52. A proton is projected in the positive x direction **W** into a region of a uniform electric field $\vec{\mathbf{E}} = (-6.00 \times 10^5)\,\hat{\mathbf{i}}\ \mathrm{N/C}$ at $t = 0$. The proton travels 7.00 cm as it comes to rest. Determine (a) the acceleration of the proton, (b) its initial speed, and (c) the time interval over which the proton comes to rest.

(a) $$|a| = \frac{qE}{m} = \frac{(1.602 \times 10^{-19}\ \mathrm{C})(6.00 \times 10^5\ \mathrm{N/C})}{1.67 \times 10^{-27}\ \mathrm{kg}} = 5.76 \times 10^{13}\ \mathrm{m/s},$$

so $\vec{\mathbf{a}} = \boxed{-5.76 \times 10^{13}\,\hat{\mathbf{i}}\ \mathrm{m/s^2}}$.

(b) $$v_f^2 = v_i^2 + 2a(x_f - x_i)$$

$$0 = v_i^2 + 2(-5.76 \times 10^{13}\ \mathrm{m/s^2})(0.070\,0\ \mathrm{m})$$

$$\vec{\mathbf{v}}_i = \boxed{2.84 \times 10^6\,\hat{\mathbf{i}}\ \mathrm{m/s}}$$

(c) $$v_f = v_i + at$$

$$0 = 2.84 \times 10^6\ \mathrm{m/s} + (-5.76 \times 10^{13}\ \mathrm{m/s^2})t \quad \rightarrow \quad t = \boxed{4.93 \times 10^{-8}\ \mathrm{s}}$$

53. An electron and a proton are each placed at rest in a uniform electric field of magnitude 520 N/C. Calculate the speed of each particle 48.0 ns after being released.

We use $v_f = v_i + at$, where $v_i = 0$, $t = 48.0 \times 10^{-9}$ s, and $a = F/m = eE/m$.

For the electron, $m = m_e = 9.11 \times 10^{-31}$ kg

and for the proton, $m = m_p = 1.67 \times 10^{-27}$ kg

The electric force on both particles is given by

$$F = eE = \left(1.60 \times 10^{-19}\ \text{C}\right)\left(5.20 \times 10^{2}\ \text{N/C}\right) = 8.32 \times 10^{-17}\ \text{N}$$

Then, for the electron,

$$v_{fe} = v_{ie} + at = 0 + \left(\frac{eE}{m_e}\right)t = \left(\frac{8.32 \times 10^{-17}\ \text{N}}{9.11 \times 10^{-31}\ \text{kg}}\right)\left(48.0 \times 10^{-9}\ \text{s}\right)$$

$$= \boxed{4.38 \times 10^{6}\ \text{m/s}}$$

and for the proton,

$$v_{fp} = v_{ip} + at = 0 + \left(\frac{eE}{m_p}\right)t = \left(\frac{8.32 \times 10^{-17}\ \text{N}}{1.67 \times 10^{-27}\ \text{kg}}\right)\left(48.0 \times 10^{-9}\ \text{s}\right)$$

$$= \boxed{2.39 \times 10^{3}\ \text{m/s}}$$

63. A line of charge starts at $x = +x_0$ and extends to positive infinity. The linear charge density is $\lambda = \lambda_0 x_0 / x$, where λ_0 is a constant. Determine the electric field at the origin.

$$\vec{\mathbf{E}} = \int d\vec{\mathbf{E}} = \int_{x_0}^{\infty} \left[\frac{k_e \lambda_0 x_0 dx \left(-\hat{\mathbf{i}}\right)}{x^3} \right] = -k_e \lambda_0 x_0 \hat{\mathbf{i}} \int_{x_0}^{\infty} x^{-3}\, dx$$

$$= -k_e \lambda_0 x_0 \hat{\mathbf{i}} \left(-\frac{1}{2x^2} \Big|_{x_0}^{\infty} \right)$$

$$= \boxed{\frac{k_e \lambda_0}{2x_0} \left(-\hat{\mathbf{i}}\right)}$$

67. M A charged cork ball of mass 1.00 g is suspended on a light string in the presence of a uniform electric field as shown in Figure P23.67. When $\vec{\mathbf{E}} = (3.00\hat{\mathbf{i}} + 5.00\hat{\mathbf{j}}) \times 10^5$ N/C, the ball is in equilibrium at $\theta = 37.0°$. Find (a) the charge on the ball and (b) the tension in the string.

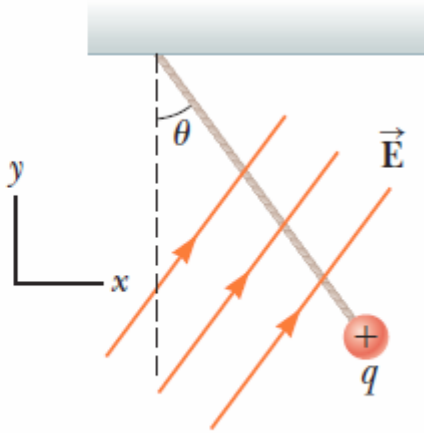
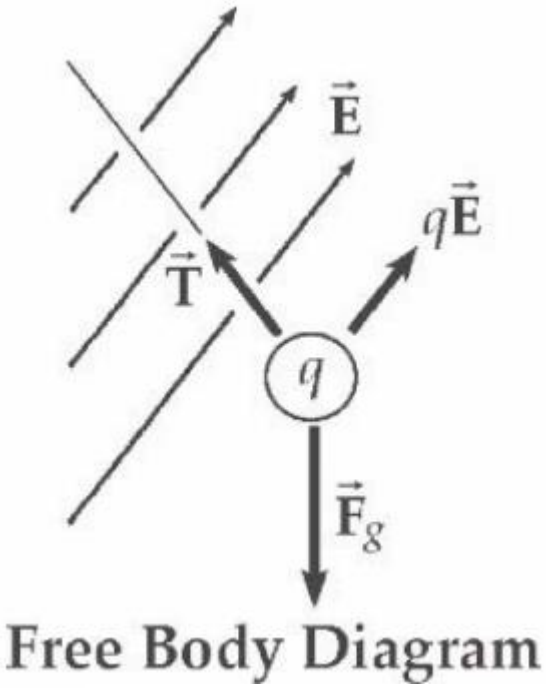


Figure P23.67
Problems 67 and 68.



$$\sum \vec{\mathbf{F}} = \vec{\mathbf{T}} + q\vec{\mathbf{E}} + \vec{\mathbf{F}}_g = 0$$

We are given

$$E_x = 3.00 \times 10^5 \text{ N/C}$$

and $$E_y = 5.00 \times 10^5 \text{ N/C}$$

$$\sum F_x = qE_x - T\sin 37.0^\circ = 0 \qquad [1]$$

$$\sum F_y = qE_y + T\cos 37.0^\circ - mg = 0 \qquad [2]$$

(a) We solve for T from equation [1]:

$$T = \frac{qE_x}{\sin 37.0^\circ}$$

and substitute into equation [2] to obtain

$$q = \frac{mg}{E_y + \dfrac{E_x}{\tan 37.0^\circ}}$$

$$= \frac{\left(1.00 \times 10^{-3}\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right)}{5.00 \times 10^5\ \text{N/C} + \left(\dfrac{3.00 \times 10^5\ \text{N/C}}{\tan 37.0^\circ}\right)}$$

$$q = \boxed{1.09 \times 10^{-8}\ \text{C}}$$

(b) Using the above result for q in equation [1], we find that the tension is

$$T = \frac{qE_x}{\sin 37.0^\circ} = \frac{\left(1.09 \times 10^{-8}\ \text{C}\right)\left(3.00 \times 10^5\ \text{N/C}\right)}{\sin 37.0^\circ}$$

$$= \boxed{5.44 \times 10^{-3}\ \text{N}}$$